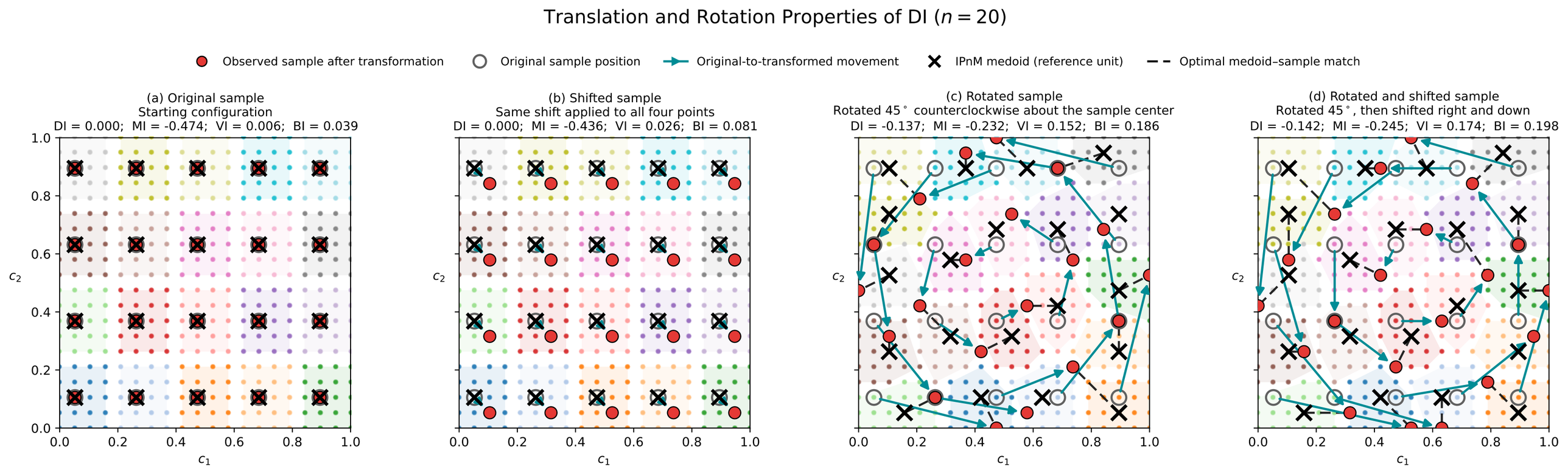

Translation and Rotation Properties of DI ($n = 20$)
Observed sample after transformation
Original sample position
Original-to-transformed movement
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Original sample
Starting configuration
DI = 0.000; MI = -0.474; VI = 0.006; BI = 0.039
(b) Shifted sample
Same shift applied to all four points
DI = 0.000; MI = -0.436; VI = 0.026; BI = 0.081
(c) Rotated sample
Rotated 45° counterclockwise about the sample center
DI = -0.137; MI = -0.232; VI = 0.152; BI = 0.186
(d) Rotated and shifted sample
Rotated 45°, then shifted right and down
DI = -0.142; MI = -0.245; VI = 0.174; BI = 0.198
c_1
c_2
0.0
0.2
0.4
0.6
0.8
1.0